

SAT Math Mastery · Week 3B

Quadratics I: Forms & Features

Time 150 minutes: 110 min solving + 40 min self-check and error log

Topics standard, vertex and factored form · completing the square · axis of symmetry and vertex · zeros and y -intercept · building a quadratic from features · maximum/minimum in context · translations of parabolas

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Choose the form that displays the quantity asked for (zeros, vertex, y -intercept).
- S2.** Convert between standard, vertex and factored form; complete the square.
- S3.** Find the axis of symmetry, vertex, and maximum or minimum value.
- S4.** Use zeros and the vertex to relate a , b , c (sum and product of zeros).
- S5.** Build a quadratic from a vertex and a point, from zeros and a point, or from a graph.
- S6.** Use symmetry: equal outputs at $x = p$ and $x = q$ place the axis at $\frac{p+q}{2}$.
- S7.** Interpret maximum/minimum, zeros and the y -intercept in context (height, revenue, area).
- S8.** Translate a parabola and track the vertex, zeros and equation.

2 · REQUIRED FACTS

- The graph of $y = ax^2 + bx + c$ is a parabola: it opens up if $a > 0$ (minimum) and down if $a < 0$ (maximum).
- The axis of symmetry is $x = -\frac{b}{2a}$, which is also the midpoint of the zeros; the vertex lies on it.
- Standard form shows the y -intercept c ; vertex form $a(x - h)^2 + k$ shows the vertex (h, k) ; factored form $a(x - r)(x - s)$ shows the zeros r, s .
- Number of real zeros: $b^2 - 4ac > 0$ gives two, $= 0$ gives one (the vertex touches the x -axis), < 0 gives none.
- Sum of zeros $= -\frac{b}{a}$, product of zeros $= \frac{c}{a}$.
- In context, the vertex is the best/worst case; the positive zero is when a launched object lands; c is the starting value.
- A negative length or time is rejected after solving.

3 · REQUIRED FORMULAS

$$y = ax^2 + bx + c, \quad y = a(x - h)^2 + k, \quad y = a(x - r)(x - s), \quad h = -\frac{b}{2a}, \quad k = c - \frac{b^2}{4a}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad \Delta = b^2 - 4ac, \quad h = \frac{r + s}{2}, \quad x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \frac{b^2}{4}$$

4 · MINI THEORY SHEET

Which form shows what?

Form	Displays	Best for
$ax^2 + bx + c$	y -intercept c , axis $-\frac{b}{2a}$	start value; quick axis
$a(x - h)^2 + k$	vertex (h, k) , max/min k	maximum or minimum, translations
$a(x - r)(x - s)$	zeros r and s	when the graph crosses the x -axis

Completing the square. $x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + \left(c - \frac{b^2}{4}\right)$. If $a \neq 1$ factor a out of the x^2 and x terms first.

Build from features. Vertex (h, k) and a point: $a(x - h)^2 + k$, solve for a . Zeros r, s and a point: $a(x - r)(x - s)$, solve for a . Never skip the point: it fixes a .

Symmetry. Equal outputs at $x = p$ and $x = q$ means the axis is $x = \frac{p+q}{2}$. A launched object returns to its starting height at twice the time of the vertex.

Context. Maximum height or profit is the y -value of the vertex; *when* it occurs is the x -value. Read which of the two is asked.

Translation. Right h / up k : replace x by $x - h$ and add k . The vertex moves by (h, k) ; a never changes.

Speed shortcuts.

Situation	Shortcut
min/max of $ax^2 + bx + c$	$x = -\frac{b}{2a}$, then substitute once
equal outputs at p and q	axis $\frac{p+q}{2}$; no equation needed
$x^2 + bx + c$ vertex height	$c - \frac{b^2}{4}$
sum / product of zeros	$-\frac{b}{a}$ and $\frac{c}{a}$ (no factoring)
exactly one zero	$b^2 = 4ac$
vertex and a point	$a = \frac{y - k}{(x - h)^2}$ (square the horizontal gap)
max of $\frac{N}{\text{quadratic}}$	minimise the positive quadratic first
rectangle with fixed fence	maximum area when the sides are split equally by cost

5 · GUIDED EXAMPLES

Example 1 — Complete the square and read the vertex

Write $y = x^2 + 10x + 7$ in vertex form and give the minimum value.

$x^2 + 10x + 7 = (x + 5)^2 - 25 + 7 = (x + 5)^2 - 18$. Vertex $(-5, -18)$; minimum value -18 .

Example 2 — Build a quadratic from zeros and a point

A parabola crosses the x -axis at -1 and 4 and passes through $(2, -12)$. Find its equation.

$y = a(x + 1)(x - 4)$. At $(2, -12)$: $-12 = a(3)(-2) = -6a \Rightarrow a = 2$. So $y = 2(x + 1)(x - 4) = 2x^2 - 6x - 8$.

Example 3 — Maximum in context

A ball's height in feet is $h(t) = -16t^2 + 64t + 5$. Find the maximum height and when it occurs.

$t = -\frac{64}{2(-16)} = 2$ seconds. $h(2) = -64 + 128 + 5 = 69$ feet. (The question decides whether the answer is 2 or 69.)

6 · PRACTICE SET — 46 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED

In the xy -plane, what are the coordinates of the vertex of the parabola $y = (x - 4)^2 - 7$?

(A) $(-4, 7)$

(B) $(4, -7)$ ✓

(C) $(-4, -7)$

(D) $(4, 7)$

1. Vertex form $y = a(x - h)^2 + k$ has vertex (h, k) .

2. Here $h = 4$ (from $x - 4$) and $k = -7$.

3. The vertex is $(4, -7)$.

Answer: (B)

2. EASY · NO CALCULATOR NEEDED

What is the positive solution to the equation $x^2 - 5x - 14 = 0$?

1. Factor: find two numbers with product -14 and sum -5 : -7 and 2 .

2. $(x - 7)(x + 2) = 0 \Rightarrow x = 7$ or $x = -2$.

3. The positive solution is 7 .

Answer: 7

3. EASY · NO CALCULATOR NEEDED

The equation $y = x^2 - 6x + 8$ defines a parabola in the xy -plane. Which of the following equivalent equations displays the x -intercepts of the parabola as constants?

(A) $y = x(x - 6) + 8$

(B) $y = (x + 2)(x + 4)$

(C) $y = (x - 2)(x - 4)$ ✓

(D) $y = (x - 3)^2 - 1$

1. Factor $x^2 - 6x + 8$: numbers with product 8 and sum -6 are -2 and -4 .

2. $y = (x - 2)(x - 4)$ is equivalent, and each factor shows an x -intercept: 2 and 4 .

3. $(x - 3)^2 - 1$ is also equivalent but shows the vertex; $(x + 2)(x + 4)$ is not equivalent.

Answer: (C)

4. EASY · NO CALCULATOR NEEDED

What is the equation of the axis of symmetry of the parabola $y = x^2 - 10x + 3$?

(A) $x = 5$ ✓

(B) $x = 10$

(C) $x = 3$

(D) $x = -5$

1. For $y = ax^2 + bx + c$ the axis is $x = -\frac{b}{2a}$.

2. $x = -\frac{-10}{2(1)} = 5$.

Answer: (A)

5.

EASY · NO CALCULATOR NEEDED

A ball's height, in feet, t seconds after it is thrown is $h(t) = -16(t - 2)^2 + 70$. What is the maximum height of the ball, in feet?

(A) 86

(B) 2

(C) 16

(D) 70 ✓

1. The vertex form gives the vertex $(2, 70)$.

2. Because $a = -16 < 0$, the vertex is the maximum, so the maximum height is 70 feet (reached at $t = 2$).

Answer: (D)

6.

MEDIUM · NO CALCULATOR NEEDED

Which of the following is equivalent to $x^2 - 8x + 3$?

(A) $(x - 8)^2 - 61$ (B) $(x + 4)^2 - 13$ (C) $(x - 4)^2 - 13$ ✓(D) $(x - 4)^2 - 5$

1. Half of -8 is -4 , and $(-4)^2 = 16$.

2. $x^2 - 8x + 3 = (x^2 - 8x + 16) - 16 + 3$.

3. $= (x - 4)^2 - 13$.

Answer: (C)

7.

MEDIUM · CALCULATOR

What is the minimum value of the function $f(x) = 3x^2 - 18x + 10$?

1. $a = 3 > 0$, so the vertex is a minimum. Axis: $x = -\frac{-18}{2(3)} = 3$.

2. $f(3) = 3(9) - 18(3) + 10 = 27 - 54 + 10 = -17$.

3. The minimum value is -17 (not 10, which is the y -intercept).

Answer: -17

8.

MEDIUM · NO CALCULATOR NEEDED

The vertex of a parabola in the xy -plane is $(2, -5)$, and the parabola passes through $(4, 3)$. Which of the following is an equation of the parabola?

(A) $y = 8(x - 2)^2 - 5$

(B) $y = 2(x - 2)^2 - 5$ ✓

(C) $y = 4(x - 2)^2 - 5$

(D) $y = 2(x + 2)^2 - 5$

1. Vertex form: $y = a(x - 2)^2 - 5$.

2. Substitute $(4, 3)$: $3 = a(2)^2 - 5 \Rightarrow 4a = 8 \Rightarrow a = 2$.

3. $y = 2(x - 2)^2 - 5$.

Answer: (B)

9.

MEDIUM · NO CALCULATOR NEEDED

A quadratic function f has zeros at $x = -3$ and $x = 5$, and the graph of $y = f(x)$ passes through $(0, -30)$. What is the value of $f(6)$?

1. $f(x) = a(x + 3)(x - 5)$.

2. $f(0) = a(3)(-5) = -15a = -30 \Rightarrow a = 2$.

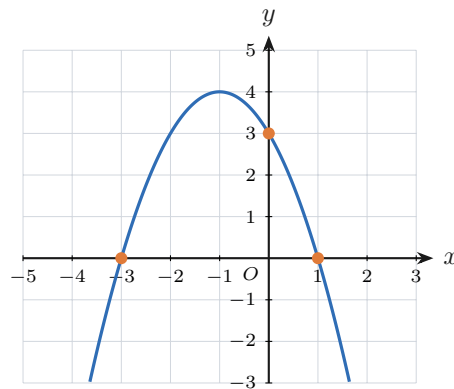
3. $f(6) = 2(9)(1) = 18$.

Answer: 18

10.

MEDIUM · DESMOS

The graph of a parabola in the xy -plane is shown. Which of the following is an equation of the parabola?



(A) $y = (x + 3)(x - 1)$

(B) $y = -(x - 3)(x + 1)$

(C) $y = -2(x + 3)(x - 1)$

(D) $y = -(x + 3)(x - 1)$ ✓

1. The x -intercepts are -3 and 1 , so $y = a(x + 3)(x - 1)$.

2. The y -intercept is 3 : $3 = a(3)(-1) \Rightarrow a = -1$.

3. $y = -(x + 3)(x - 1)$.

Answer: (D)

11.

MEDIUM · NO CALCULATOR NEEDED

A company's weekly revenue, in dollars, from selling a product at a price of p dollars is $R(p) = -4p^2 + 120p$. Which price maximizes the weekly revenue, in dollars?

(A) 15 ✓

(B) 30

(C) 450

(D) 900

1. The maximum is at $p = -\frac{120}{2(-4)} = 15$.

2. The maximum revenue itself, $R(15) = 900$, is a different quantity; the question asks for the price.

Answer: (A)

12.

MEDIUM · NO CALCULATOR NEEDED

The function $f(x) = x^2 + bx - 21$ has a zero at $x = 3$, where b is a constant. What is the x -coordinate of the vertex of the graph of $y = f(x)$?

1. $f(3) = 0$: $9 + 3b - 21 = 0 \Rightarrow b = 4$.

2. Vertex: $x = -\frac{b}{2} = -2$.

3. (Check: $x^2 + 4x - 21 = (x - 3)(x + 7)$, zeros 3 and -7 , midpoint -2 .)

Answer: -2

13.

MEDIUM · NO CALCULATOR NEEDED

The graph of $y = x^2 - 6x + 5$ is translated 2 units to the left. Which of the following is an equation of the translated graph?

(A) $y = x^2 - 6x + 7$

(B) $y = x^2 - 2x - 3$ ✓

(C) $y = x^2 - 10x + 21$

(D) $y = x^2 - 6x + 3$

1. Moving left 2 replaces x with $x + 2$: $y = (x + 2)^2 - 6(x + 2) + 5$.

2. $= x^2 + 4x + 4 - 6x - 12 + 5 = x^2 - 2x - 3$.

Answer: (B)

14.

HARD · NO CALCULATOR NEEDED

The function $f(x) = x^2 - bx + c$ has its vertex at $(4, -7)$, where b and c are constants. What is the value of c ?

(A) 16

(B) -23

(C) 9 ✓

(D) 23

1. Vertex x -coordinate: $\frac{b}{2} = 4 \Rightarrow b = 8$.

2. $f(4) = 16 - 32 + c = -7 \Rightarrow c = 9$.

3. (Check via vertex form: $(x - 4)^2 - 7 = x^2 - 8x + 9$.)

Answer: (C)

15.

HARD · NO CALCULATOR NEEDED

For how many integer values of k with $-10 \leq k \leq 10$ does the equation $x^2 + kx + 16 = 0$ have no real solutions?

1. No real solutions means the discriminant is negative: $k^2 - 4(1)(16) < 0 \Rightarrow k^2 < 64$.
2. So $-8 < k < 8$.
3. The integers $-7, -6, \dots, 7$: that is 15 values (all inside $-10 \leq k \leq 10$).

Answer: 15

16.

HARD · CALCULATOR

A ball is thrown upward from a platform. Its height, in feet, t seconds after the throw is $h(t) = -16t^2 + bt + c$, where b and c are constants. The ball is at the same height at $t = 1$ and $t = 4$, and its height at $t = 0$ is 20 feet. What is the maximum height, in feet, of the ball?

1. Equal heights at $t = 1$ and $t = 4$ put the axis at $t = \frac{1+4}{2} = 2.5$.
2. $-\frac{b}{2(-16)} = 2.5 \Rightarrow b = 80$. Also $c = h(0) = 20$.
3. $h(2.5) = -16(6.25) + 80(2.5) + 20 = -100 + 200 + 20 = 120$.

Answer: 120

17.

HARD · NO CALCULATOR NEEDED

The graph of $y = -2x^2 + 12x - 10$ is a parabola in the xy -plane. Which of the following statements must be true?

- I. The parabola opens upward.
II. The vertex of the parabola is $(3, 8)$.
III. The x -intercepts of the parabola are 1 and 5.

(A) II and III only ✓

(B) I, II, and III

(C) I only

(D) II only

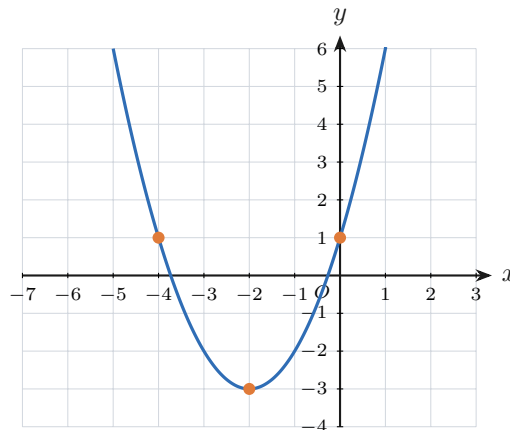
1. I: $a = -2 < 0$, so the parabola opens downward. False.
2. II: $x = -\frac{12}{2(-2)} = 3$ and $y = -2(9) + 36 - 10 = 8$. True.
3. III: $-2x^2 + 12x - 10 = -2(x-1)(x-5)$, zeros 1 and 5. True.
4. II and III only.

Answer: (A)

18.

HARD · DESMOS

The graph of $y = f(x)$ is shown in the xy -plane, where $f(x) = ax^2 + bx + c$ and a , b , and c are constants. What is the value of b ?



(A) 6

(B) -4

(C) 2

(D) 4 ✓

1. The vertex is $(-2, -3)$: $f(x) = a(x + 2)^2 - 3$.
2. The graph passes through $(0, 1)$: $1 = 4a - 3 \Rightarrow a = 1$.
3. $f(x) = (x + 2)^2 - 3 = x^2 + 4x + 1$, so $b = 4$.

Answer: (D)

19.

HARD · CALCULATOR

A farmer uses 120 feet of fencing to enclose a rectangular pen. One side of the pen is a long barn wall and needs no fencing, so the fencing covers the other three sides. What is the maximum possible area of the pen, in square feet?

(A) 7200

(B) 900

(C) 1800 ✓

(D) 3600

1. Let the two sides perpendicular to the wall be w ; the side parallel to the wall is $120 - 2w$.
2. $A = w(120 - 2w) = -2w^2 + 120w$.
3. Maximum at $w = -\frac{120}{2(-2)} = 30$, giving length 60.
4. $A = 30 \cdot 60 = 1800$ square feet.

Answer: (C)

20.

HARD · NO CALCULATOR NEEDED

The graph of $f(x) = x^2 - 4x + 7$ is translated 5 units to the right and 3 units down to produce the graph of $y = g(x)$. What is the y -intercept of the graph of g ?

1. Right 5 and down 3: $g(x) = f(x - 5) - 3$.
2. y -intercept: $g(0) = f(-5) - 3$.
3. $f(-5) = 25 + 20 + 7 = 52$, so $g(0) = 52 - 3 = 49$.
4. (Check with vertices: f has vertex $(2, 3)$, g has vertex $(7, 0)$ so $g(x) = (x - 7)^2 = x^2 - 14x + 49$.)

Answer: 49**21.**

HARD · NO CALCULATOR NEEDED

The expression $2x^2 - 12x + 7$ can be written in the form $2(x - h)^2 + k$, where h and k are constants. What is the value of $h + k$?

1. Factor 2 from the x -terms: $2(x^2 - 6x) + 7$.
2. Complete the square: $x^2 - 6x = (x - 3)^2 - 9$, so $2(x - 3)^2 - 18 + 7$ after distributing the 2.
3. $= 2(x - 3)^2 - 11$, so $h = 3$ and $k = -11$.
4. $h + k = 3 + (-11) = -8$.

Answer: -8**22.**

HARD · NO CALCULATOR NEEDED

In the xy -plane, the graph of $y = x^2 - 8x + c$ has two distinct x -intercepts. Which of the following could *not* be the value of c ?

(A) 15

(B) 16 ✓

(C) -3

(D) 12

1. Two distinct x -intercepts need $\Delta = (-8)^2 - 4(1)c > 0 \Rightarrow 64 - 4c > 0 \Rightarrow c < 16$.
2. -3, 12, and 15 are all less than 16.
3. $c = 16$ gives $\Delta = 0$ (one intercept, since $x^2 - 8x + 16 = (x - 4)^2$), so it could not be the value.

Answer: (B)

23.

HARD · NO CALCULATOR NEEDED

The two solutions of the equation $x^2 - 9x + k = 0$ differ by 3, where k is a constant. What is the value of k ?

(A) 20

(B) 27

(C) 9

(D) 18 ✓

1. Let the solutions be r and $r + 3$. Their sum is 9: $2r + 3 = 9 \Rightarrow r = 3$.

2. The solutions are 3 and 6, so $k = 3 \cdot 6 = 18$.

3. (Check: $x^2 - 9x + 18 = (x - 3)(x - 6)$.)

Answer: (D)

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

A parabola in the xy -plane has x -intercepts at $(1, 0)$ and $(5, 0)$, and its maximum value is 12. If the equation of the parabola is $y = ax^2 + bx + c$, what is the value of a ?

1. Write $y = a(x - 1)(x - 5)$. The axis is $x = \frac{1 + 5}{2} = 3$.

2. The maximum is at the vertex: $y(3) = a(2)(-2) = -4a = 12$.

3. $a = -3$. (Negative, consistent with a maximum.)

Answer: -3

25.

SAT CHALLENGE · CALCULATOR

A ball is launched upward from a cliff. Its height, in feet, t seconds after launch is $h(t) = -16t^2 + vt + s$, where v and s are constants. The ball is 100 feet above the ground at $t = 1$ and again at $t = 5$. What is the maximum height, in feet, of the ball?

1. Equal heights at $t = 1$ and $t = 5$ place the axis at $t = 3$: $\frac{v}{32} = 3 \Rightarrow v = 96$.

2. $h(1) = -16 + 96 + s = 100 \Rightarrow s = 20$.

3. $h(3) = -16(9) + 96(3) + 20 = -144 + 288 + 20 = 164$.

4. Maximum height 164 feet. (Check: $h(5) = -400 + 480 + 20 = 100$.)

Answer: 164

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

The graph of $y = x^2 + kx + 9$ touches the x -axis at exactly one point, and that point has a negative x -coordinate. What is the value of k ?

(A) 6 ✓

(B) -6

(C) -3

(D) 3

1. Touching at one point: discriminant 0. $k^2 - 4(1)(9) = 0 \Rightarrow k^2 = 36 \Rightarrow k = \pm 6$.

2. The touching point is the vertex, $x = -\frac{k}{2}$. For it to be negative, $k > 0$.

3. $k = 6$. (Then $y = (x + 3)^2$ touches at $x = -3$; $k = -6$ gives $x = 3$.)

Answer: (A)

27.

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function $f(x) = x^2 - bx + c$ has two zeros whose sum is 12 and whose product is 27. The function g is defined by $g(x) = f(2x)$. What is the sum of the zeros of g ?

1. Zeros of f : sum 12, product 27, so they are 3 and 9 (since $(x - 3)(x - 9) = x^2 - 12x + 27$).
2. $g(x) = 0$ when $f(2x) = 0$, i.e. $2x = 3$ or $2x = 9$.
3. $x = \frac{3}{2}$ or $x = \frac{9}{2}$.
4. Sum $= \frac{3}{2} + \frac{9}{2} = 6$.

Answer: 6**28.**

SAT CHALLENGE · NO CALCULATOR NEEDED

What is the maximum value of $\frac{20}{x^2 - 6x + 13}$ over all real numbers x ?

1. The denominator is always positive: $x^2 - 6x + 13 = (x - 3)^2 + 4 \geq 4$.
2. The fraction is largest when the denominator is smallest, at $x = 3$, where the denominator equals 4.
3. Maximum value $= \frac{20}{4} = 5$.

Answer: 5

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

The parabola $y = x^2 - 2px + q$ passes through the point $(0, 6)$, and its vertex lies on the line $y = x - 6$. If $p > 0$, what is the value of p ?

1. $(0, 6)$ gives $q = 6$. Vertex form: $y = (x - p)^2 + 6 - p^2$, vertex $(p, 6 - p^2)$.
2. On the line $y = x - 6$: $6 - p^2 = p - 6 \Rightarrow p^2 + p - 12 = 0 \Rightarrow (p + 4)(p - 3) = 0$.
3. $p = 3$ or $p = -4$; the condition $p > 0$ gives $p = 3$.
4. (Check: vertex $(3, -3)$, and $-3 = 3 - 6$.)

Answer: 3**30.**

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the parabola $y = -(x - 2)^2 + 9$ has its vertex at point V and crosses the x -axis at points A and B . What is the area of triangle VAB ?

(A) 36

(B) 27 ✓

(C) 54

(D) 18

1. Vertex: $V = (2, 9)$.
2. x -intercepts: $(x - 2)^2 = 9 \Rightarrow x = -1$ or $x = 5$, so $AB = 6$.
3. The height from V to the x -axis is 9.
4. Area = $\frac{1}{2}(6)(9) = 27$.

Answer: (B)

31.

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function $f(x) = ax^2 + bx + c$, where a , b and c are constants, has two distinct real zeros. It is known that $f(2) = f(8)$, $f(0) < 0$ and $f(5) > 0$. Consider the following statements.

- I. $b > 0$
- II. The product of the two zeros is greater than 25.
- III. Both zeros are positive.

Which of the statements must be true?

- (A) I only
- (B) II and III only
- (C) I and III only ✓**
- (D) I, II, and III

1. $f(2) = f(8)$ puts the axis at $x = 5$, so $-\frac{b}{2a} = 5$ and $b = -10a$. The zeros sum to $-\frac{b}{a} = 10$.
2. $f(5) > 0$ says the vertex is above the x -axis. A parabola with two zeros and a vertex above the axis opens downward, so $a < 0$.
3. I: $b = -10a > 0$. True.
4. II: zeros r, s with $r + s = 10$ and $r \neq s$ give $rs < \left(\frac{r+s}{2}\right)^2 = 25$. False (the product is always less than 25).
5. III: $f(0) = c < 0$ and $a < 0$, so $rs = \frac{c}{a} > 0$; with $r + s = 10 > 0$ both zeros are positive. True.
6. Only I and III must be true.

Answer: (C)

32.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined by $f(x) = ax^2 + bx + c$, where a , b , and c are constants. The graph of $y = f(x)$ in the xy -plane passes through $(1, 8)$ and $(-1, 2)$ and has exactly one x -intercept. What is the greatest possible value of a ?

- (A) $\frac{5}{2}$
- (B) 9
- (C) $\frac{9}{2}$ ✓**
- (D) $\frac{1}{2}$

1. $f(1) = a + b + c = 8$ and $f(-1) = a - b + c = 2$. Subtract: $2b = 6 \Rightarrow b = 3$. Add: $a + c = 5 \Rightarrow c = 5 - a$.
2. Exactly one x -intercept: $b^2 - 4ac = 0 \Rightarrow 9 = 4a(5 - a)$.
3. $4a^2 - 20a + 9 = 0 \Rightarrow (2a - 1)(2a - 9) = 0 \Rightarrow a = \frac{1}{2}$ or $a = \frac{9}{2}$.
4. The greatest possible value is $\frac{9}{2}$.

Answer: (C)

33.

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function f has its vertex at $(3, -2)$, and $f(5) = 6$. The function g is defined by $g(x) = \frac{3}{4}f\left(\frac{1}{2}x - 1\right)$. The graph of $y = f(x)$ has y -intercept $(0, p)$ and the graph of $y = g(x)$ has y -intercept $(0, q)$. What is the value of $q - p$?

(A) $\frac{13}{2}$ ✓

(B) -4

(C) $\frac{45}{2}$

(D) $-\frac{23}{2}$

1. Vertex form: $f(x) = a(x - 3)^2 - 2$. From $f(5) = 6$: $4a - 2 = 6 \Rightarrow a = 2$, so $f(x) = 2(x - 3)^2 - 2$.

2. $p = f(0) = 2(9) - 2 = 16$.

3. $q = g(0) = \frac{3}{4}f\left(\frac{1}{2} \cdot 0 - 1\right) = \frac{3}{4}f(-1)$, and $f(-1) = 2(16) - 2 = 30$, so $q = \frac{45}{2}$.

4. $q - p = \frac{45}{2} - 16 = \frac{13}{2}$.

Answer: (A)

34.

SAT CHALLENGE · CALCULATOR

The altitude h , in meters, of a small rocket t seconds after launch is modeled by $h(t) = at^2 + bt + c$, where a , b , and c are constants. At launch the altitude is 3 meters; 2 seconds after launch it is 43 meters; and 5 seconds after launch it is 28 meters. According to the model, what is the altitude, in meters, 3 seconds after launch?

1. Launch altitude: $h(0) = c = 3$.

2. $h(2) = 4a + 2b + 3 = 43 \Rightarrow 2a + b = 20$ (divide by 2).

3. $h(5) = 25a + 5b + 3 = 28 \Rightarrow 5a + b = 5$ (divide by 5).

4. Subtract: $3a = -15 \Rightarrow a = -5$; then $b = 20 - 2(-5) = 30$.

5. $h(3) = -5(9) + 30(3) + 3 = -45 + 90 + 3 = 48$.

Answer: 48

35.

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function f has zeros at $x = -2$ and $x = 5$, and $f(1) = -24$. The function h is defined by $h(x) = f(x + 2) - 3f(-x)$. What is the value of $h(1)$?

1. $f(x) = a(x + 2)(x - 5)$; $f(1) = a(3)(-4) = -24 \Rightarrow a = 2$, so $f(x) = 2(x + 2)(x - 5)$.
2. Substitute $x = 1$ first: $h(1) = f(3) - 3f(-1)$.
3. $f(3) = 2(5)(-2) = -20$ and $f(-1) = 2(1)(-6) = -12$.
4. $h(1) = -20 - 3(-12) = -20 + 36 = 16$.

Answer: 16**36.**

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function f has an axis of symmetry at $x = 2$, and its graph passes through $(3, -3)$ and $(5, 13)$. The functions g and h are defined by $g(x) = \frac{x-6}{2}$ and $h(x) = 3 - f(g(x))$. What is the y -coordinate of the y -intercept of the graph of $y = h(x)$?

1. Vertex form with axis $x = 2$: $f(x) = a(x - 2)^2 + k$.
2. $f(3) = a + k = -3$ and $f(5) = 9a + k = 13$. Subtract: $8a = 16 \Rightarrow a = 2$, $k = -5$.
3. $f(x) = 2(x - 2)^2 - 5$. The y -intercept of h is $h(0)$.
4. Inner value first: $g(0) = -3$. Then $f(-3) = 2(-5)^2 - 5 = 45$.
5. $h(0) = 3 - 45 = -42$.

Answer: -42

37.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$-2(x-5)^2 + k = m(x-1)(x-r)$$

In the equation above, k , m , and r are constants. The equation is true for all real values of x . What is the value of k ?

(A) 50

(B) -32

(C) 18

(D) 32 ✓

1. Leading coefficients: left side $-2x^2 + \dots$, right side $mx^2 + \dots$, so $m = -2$.
2. The axis of the left side is $x = 5$; the right side has roots 1 and r , so $\frac{1+r}{2} = 5$ and $r = 9$.
3. Substitute the root $x = 1$, where the right side is 0: $-2(1-5)^2 + k = 0 \Rightarrow -32 + k = 0 \Rightarrow k = 32$.
4. (Check: $-2(x-5)^2 + 32 = -2x^2 + 20x - 18 = -2(x-1)(x-9)$.)

Answer: (D)

38.

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function f has a zero at $x = 2$. The graph of $y = f(x)$ in the xy -plane passes through the points $(0, -16)$ and $(5, 54)$. What is the minimum value of f ?

(A) -9

(B) -1

(C) -18 ✓

(D) -16

1. Let the other zero be s and write $f(x) = a(x-2)(x-s)$.
2. $f(0) = a(-2)(-s) = 2as = -16$ and $f(5) = a(3)(5-s) = 54$.
3. Divide the second equation by the first to cancel a : $\frac{3(5-s)}{2s} = \frac{54}{-16} = -\frac{27}{8}$. Then $24(5-s) = -54s$, so $120 = -30s$ and $s = -4$.
4. Then $2a(-4) = -16 \Rightarrow a = 2 > 0$, so the parabola opens upward and has a minimum.
5. The vertex is at $x = \frac{2+(-4)}{2} = -1$: $f(-1) = 2(-3)(3) = -18$.
6. The minimum value is -18 . (The options -16 and -1 are $f(0)$ and the vertex's x -coordinate.)

Answer: (C)

39.

SAT CHALLENGE · NO CALCULATOR NEEDED

The graph of the quadratic function f in the xy -plane passes through the points $(0, 1)$, $(1, -2)$ and $(3, 4)$. The graph of $y = g(x)$ is obtained by translating the graph of $y = f(x)$ 3 units to the right and 5 units down. What is the y -coordinate of the y -intercept of the graph of g ?

(A) -1 **(B) 29 ✓**

(C) 34

(D) 39

1. Let $f(x) = ax^2 + bx + c$. From $(0, 1)$: $c = 1$.
2. From $(1, -2)$: $a + b + 1 = -2 \Rightarrow a + b = -3$. From $(3, 4)$: $9a + 3b + 1 = 4 \Rightarrow 3a + b = 1$.
3. Subtract: $2a = 4$, so $a = 2$ and $b = -5$. Thus $f(x) = 2x^2 - 5x + 1$.
4. The translation gives $g(x) = f(x - 3) - 5$, so the y -intercept is $g(0) = f(-3) - 5$.
5. $f(-3) = 2(9) + 15 + 1 = 34$, so $g(0) = 34 - 5 = 29$.
6. The answer is 29. (The options 34 and 39 forget the shift down or reverse it; -1 uses $f(3) - 5$, the wrong direction of the horizontal shift.)

Answer: (B)

40.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function $f(x) = 2(3x + 1 - \sqrt{5})(3x + 1 + \sqrt{5})$ can be written as $f(x) = ax^2 + bx + c$, where a , b and c are constants. What is the value of $\frac{b+c}{a}$?

1. The factors are conjugates of the form $(u-v)(u+v)$ with $u = 3x+1$ and $v = \sqrt{5}$, so the product is $u^2 - v^2 = (3x+1)^2 - 5 = 9x^2 + 6x - 4$.
2. Keep the leading factor 2: $f(x) = 2(9x^2 + 6x - 4) = 18x^2 + 12x - 8$. So $a = 18$, $b = 12$, $c = -8$.
3. $\frac{b+c}{a} = \frac{12-8}{18} = \frac{4}{18} = \frac{2}{9}$.
4. (Shortcut: the leading factor 2 multiplies every coefficient, so it cancels in any ratio: $\frac{b+c}{a} = \frac{6-4}{9} = \frac{2}{9}$.) Enter $2/9$.

Answer: 2/9

41.

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function $h(x) = ax^2 + bx + c$, where a , b and c are constants, satisfies $h(-3) = h(1)$, $h(0) > h(-1)$ and $h(-1) > 0$. Consider the following statements.

- I. $b > 0$
- II. $c > 0$
- III. The equation $h(x) = 0$ has two distinct real solutions.

Which of the statements must be true?

(A) I only

(B) I and II only ✓

(C) II and III only

(D) I, II, and III

1. $h(-3) = h(1)$ puts the axis at $x = \frac{-3+1}{2} = -1$, so the vertex is $(-1, h(-1))$ and $-\frac{b}{2a} = -1$, i.e. $b = 2a$.
2. $h(0) = c$ is the y -intercept and $h(-1)$ is the vertex height. Since $h(0) > h(-1)$ the vertex is the lowest point, so the parabola opens upward: $a > 0$.
3. I: $b = 2a > 0$. True.
4. II: $c = h(0) > h(-1) > 0$. True.
5. III: the minimum value $h(-1)$ is positive and the parabola opens upward, so the graph never reaches the x -axis: no real solutions. False.
6. I and II only.

Answer: (B)

42.

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function f has its vertex at $(3, 4)$. The function g is defined by $g(x) = -2f(x+1) + 7$, and the graph of $y = g(x)$ passes through the point $(4, 2)$. What is the value of $f(9) - f(2)$?

1. Vertex form: $f(x) = a(x-3)^2 + 4$.
2. Use the point on g : $g(4) = -2f(5) + 7 = 2 \Rightarrow f(5) = \frac{5}{2}$. (The shift gives $f(4+1) = f(5)$.)
3. $f(5) = a(2)^2 + 4 = \frac{5}{2} \Rightarrow 4a = -\frac{3}{2} \Rightarrow a = -\frac{3}{8}$.
4. $f(9) = -\frac{3}{8}(36) + 4 = -\frac{27}{2} + 4 = -\frac{19}{2}$ and $f(2) = -\frac{3}{8}(1) + 4 = \frac{29}{8}$.
5. $f(9) - f(2) = -\frac{76}{8} - \frac{29}{8} = -\frac{105}{8}$. (Shortcut: the $+4$ cancels, so the difference is $a(36-1) = 35 \cdot \left(-\frac{3}{8}\right) = -\frac{105}{8}$.)

Answer: $-\frac{105}{8}$

43.

SAT CHALLENGE · NO CALCULATOR NEEDED

A rocket's height, in feet, t seconds after launch is modeled by $h(t) = -16t^2 + bt + c$, where b and c are constants. The rocket is launched from a height of at most 94 feet, and the maximum height it reaches is 130 feet. For how many seconds is the rocket at a height of at least 94 feet?

1. Write the function in vertex form: $h(t) = -16(t - p)^2 + 130$, where p is the time of the maximum.
2. Height at least 94: $-16(t - p)^2 + 130 \geq 94 \Rightarrow (t - p)^2 \leq \frac{36}{16} = \frac{9}{4}$.
3. So $|t - p| \leq \frac{3}{2}$, i.e. $p - \frac{3}{2} \leq t \leq p + \frac{3}{2}$. The value of p does not matter (launch height at most 94 keeps the whole interval after $t = 0$).
4. Duration $= 2 \cdot \frac{3}{2} = 3$ seconds. (Check with $p = \frac{5}{2}$: $h(1) = h(4) = 94$.)

Answer: 3**44.**

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function $f(x) = 2x^2 + bx + c$ has $(x - 9)$ as a factor. The function g is defined by $g(x) = f(x + 3)$, and the graph of g in the xy -plane has its vertex at $x = 1$. What is the other zero of f ?

1. The graph of $g(x) = f(x + 3)$ is the graph of f shifted 3 units left. So the vertex of f is at $x = 1 + 3 = 4$.
2. The vertex lies midway between the zeros of f . One zero is 9, so the other is $2(4) - 9 = -1$.
3. (Check: $-\frac{b}{4} = 4 \Rightarrow b = -16$; $f(9) = 0 \Rightarrow c = -162 + 144 = -18$; $f(x) = 2(x - 9)(x + 1)$ and g has vertex at $x = 4 - 3 = 1$.)

Answer: -1

45.

SAT CHALLENGE · NO CALCULATOR NEEDED

The quadratic function f satisfies $f(-1) = f(7)$, $f(0) = -4$ and $f(4) = 12$. What is the value of $f(10)$?

(A) 0

(B) 98

(C) -98

(D) -84 ✓

1. Equal outputs at $x = -1$ and $x = 7$ put the axis of symmetry at $x = \frac{-1+7}{2} = 3$. Write $f(x) = a(x-3)^2 + k$.

2. $f(0) = -4$: $9a + k = -4$. $f(4) = 12$: $a + k = 12$.

3. Subtract: $8a = -16$, so $a = -2$, and then $k = 14$. So $f(x) = -2(x-3)^2 + 14$.

4. $f(10) = -2(7)^2 + 14 = -98 + 14 = -84$.

5. (Options -98 drops k ; 0 forgets to square.)

Answer: (D)

46.

SAT CHALLENGE · NO CALCULATOR NEEDED

A water balloon is launched upward from a rooftop. Its height above the ground, in feet, t seconds after launch is modeled by a quadratic function h . The maximum height of 125 feet is reached at a time between $t = 1$ and $t = 6$, and $h(1) = 80$ and $h(6) = 105$. What is the height, in feet, of the balloon at the moment of launch?

(A) 45 ✓

(B) 75

(C) 37

(D) 80

1. Write $h(t) = 125 - a(t-v)^2$, where v is the time of maximum height. The drops below the maximum are $125 - 80 = 45$ at $t = 1$ and $125 - 105 = 20$ at $t = 6$.

2. Divide: $\frac{a(v-1)^2}{a(6-v)^2} = \frac{45}{20} = \frac{9}{4}$, so $\frac{v-1}{6-v} = \frac{3}{2}$ (the maximum lies between the two times).

3. $2v - 2 = 18 - 3v \Rightarrow v = 4$. Then $45 = a(3)^2 \Rightarrow a = 5$.

4. $h(0) = 125 - 5(4)^2 = 125 - 80 = 45$.

5. (Option 75 continues the average rate of change from $t = 1$ to $t = 6$ backwards; 80 is $h(1)$; 37 places the vertex halfway at $t = 3.5$.)

Answer: (A)